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Unit – 1 : Session – 8 : SLO – 1 | Binary arithmetic and BCD operations

■ I. Match the Following

1. Sign Magnitude representation — b) Separate bit for sign, remaining for magnitude
2. 1's Complement — e) Invert all bits of the subtrahend
3. 2's Complement — c) Invert all bits and add 1 to LSB
4. Binary Multiplication Rule — d) $1 \times 1 = 1$ (partial-product/shift-add rule)
5. BCD Addition — a) Add 0110 (6) for correction if sum > 9

■ II. Fill in the Blanks

1. sign
2. inverting (complementing)
3. 1
4. 1
5. 0110

■ III. Activity / Describe and Perform

1. Sign Magnitude — Addition: same signs $\rightarrow$ add magnitudes, keep sign ($+3=0011$, $+2=0010 \rightarrow 0101=+5$). Subtraction $(+5)-(+3)$: compare magnitudes, subtract smaller from larger ($101-011=010$), keep sign of larger $\rightarrow 0010 = +2$.
2. 1's Complement — Addition: add all bits incl. sign, add end-around carry. Subtraction 7-2: $0111 + 1$'s of 0010 (1101) = $1\ 0100 \rightarrow \text{EAC} \rightarrow 0100+1 = 0101 = +5$.
3. 2's Complement — Subtraction 7-2: $0111 + 2$'s of 0010 (1110) = $1\ 0101 \rightarrow$ discard carry $\rightarrow 0101 = +5$. No carry $\rightarrow$ negative result, take 2's complement for magnitude.
4. Multiplication 1011×101 (11×5): partial products by shift-and-add $\rightarrow 0110111$ (55). Division $1010 \div 10$ ($10 \div 2$): repeated shift-and-subtract (long division) $\rightarrow$ quotient 0101 (5), remainder 0.
5. BCD: each decimal digit $\rightarrow$ 4-bit code. Example $8+5$: $1000+0101 = 1101 (>1001) \rightarrow +0110 \rightarrow 1\ 0011 \rightarrow 0001\ 0011 = 13$.